

PDF, TIFF, or image

source order

Page preparation
300 dpi render, EXIF transpose, page and pixel limits

page image

Orientation guard
docTR proposal and Tesseract OSD, up to four scored views

upright view

Routed local view
dark-frame crop, Otsu against Sauvola

Tiny-text tiles
three overlapping bands, fused rather than replaced

Selective wide-band recovery
gated 3× reread of shallow low-confidence bands

Recognizer
Tesseract floor, Nemotron OCR v2, Granite Docling

positioned literal text

Ordered evidence IR: TextRegion[]
kind, literal text, pixel box, reading order, provider, confidence, alternatives, resolution, structure

alternative

Handwriting sidcar
user-selected crop,
Phi-4 decoder LoRA

regions in

Controls
geometric squares,
label association

Tables
TATR geometry,
per-cell fusion

Evidence risk
confidence, height,
coverage reasons

*each stage is a
RegionStage:
enrich, never
overwrite*

enriched regions

Verification and routing
literal risks, candidate disagreement, structure checks, retained failures

route and evidence

accept_local or review
schema v2 JSON, Markdown, page overlays, failure view